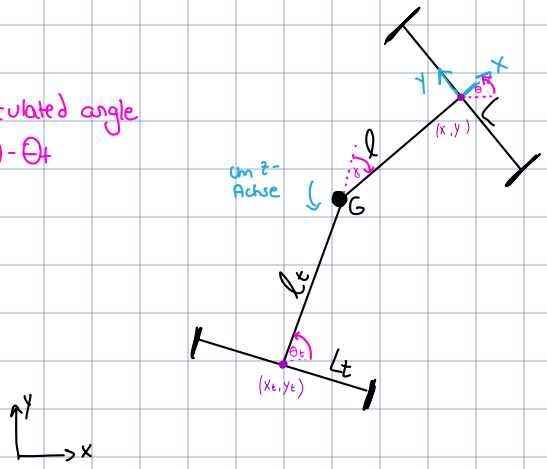
"Modeling, Control and Path Planning for an Articulated Vehicle" (Thaler Mayl)



vorderer Teil (Roboter, aber Räder drehen links und rechts gleich schnell)

- M, I, L

hinterer Teil (Anhänger)

- Mt, It, Lt

- Annahme: (x, y), (xt, yt) im jeweiligen Schwerpunkt

- no lateral slip (vgl. Mayrs Diss S. 23 (2.35))

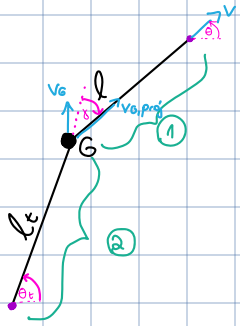
$$\dot{x}\sin\theta - \dot{y}\cos\theta = 0$$

$$\dot{x}_t \sin\theta_t - \dot{y}_t \cos\theta_t = 0$$

- $\dot{x} = v\cos\theta$

$\dot{y} = v\sin\theta$

-



(1) Geschwindigkeit des Gelenks bzgl. Roboter

$$v_G = v + \omega \times r_G$$

(in 2D $\omega \times \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} = \begin{pmatrix} -\omega b \\ \omega a \end{pmatrix}$)

$$= \begin{pmatrix} \cos\theta v \\ \sin\theta v \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot{\theta} \end{pmatrix} \times \left(\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -l \\ 0 \\ 0 \end{pmatrix} \right)$$

$$= \begin{pmatrix} \cos\theta v \\ \sin\theta v \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot{\theta} \end{pmatrix} \times \begin{pmatrix} -\cos\theta\, l \\ -\sin\theta\, l \\ 0 \end{pmatrix} = \begin{pmatrix} \cos\theta v + l\dot{\theta}\sin\theta \\ \sin\theta v - l\dot{\theta}\cos\theta \\ 0 \end{pmatrix}$$

(2) Geschwindigkeit des Gelenks bzgl. trailer

$$v_G = \ldots = \begin{pmatrix} \cos\theta_t v_t - l_t\dot{\theta}_t \sin\theta_t \\ \sin\theta_t v_t + l_t\dot{\theta}_t \cos\theta_t \\ 0 \end{pmatrix} \quad 1$$

- Geschwindigkeiten auf Geschwindigkeitsrichtung v_{dir} von Roboter projizieren:

① $|v_{G,proj}| = v_G^T v_{dir} = v_G^T \begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix} = v + l\dot{\theta}\sin\theta\cos\theta - l\dot{\theta}\cos\theta\sin\theta = v$

② $|v_{G,proj}| = v_G^T v_{dir} = v_G^T \begin{pmatrix} \cos\alpha \\ \sin\alpha \\ 0 \end{pmatrix} = (\cos\theta_t v_t - l_t\dot{\theta}_t \sin\theta_t)\cos\theta + (\sin\theta_t v_t + l_t\dot{\theta}_t \cos\theta_t)\sin\theta$

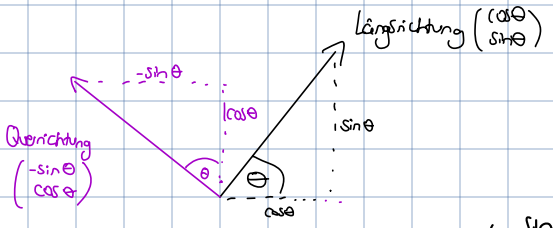
$\cos a\cos b + \sin a\sin b = \cos(a-b)$

$\sin a\cos b - \cos a\sin b = \sin(a-b)$

$= v_t \cos(\underbrace{\theta - \theta_t}_{=\gamma}) + l_t\dot{\theta}_t \sin(\underbrace{\theta - \theta_t}_{=\gamma})$

$\Rightarrow \quad v = v_t\cos\gamma + \dot{\theta}_t l_t \sin\gamma \qquad (2.5)$

- Geschwindigkeiten auf Querrichtung projizieren



① $v_{G,proj} = v_G^T v_{quer} = v_G^T \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix} = (\cos\theta v + l\dot{\theta}\sin\theta)(-\sin\theta) + (\sin\theta v - l\theta\cos\theta)\cos\theta$

$= -l\dot{\theta}(\sin^2\theta + \cos^2\theta) = -l\dot{\theta}$

② $v_{G,proj} = v_G^T v_{quer} = v_G^T \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$

$= (\cos\theta_t v_t - l_t\dot{\theta}_t \sin\theta_t)(-\sin\theta) + (\sin\theta_t v_t + l_t\dot{\theta}_t \cos\theta_t)\cos\theta$

$\cos a\cos b + \sin a\sin b = \cos(a-b)$

$\sin a\cos b - \cos a\sin b = \sin(a-b)$

$= v_t \sin(\theta_t - \theta) + l_t\dot{\theta}_t \cos(\theta - \theta_t)$

$\Rightarrow \quad -l\dot{\theta} = -v_t \sin(\underbrace{\theta - \theta_t}_{=\gamma}) + l_t\dot{\theta}_t \cos(\underbrace{\theta - \theta_t}_{=\gamma})$

$\Leftrightarrow \quad v_t\sin(\gamma) = l\dot{\theta} + l_t\dot{\theta}_t\cos(\gamma) \qquad (2.6)$

$v = v_t\cos\gamma + \dot{\theta}_t l_t\sin\gamma \qquad (2.5)$

(2.6) $\cdot \cos(\gamma)$

$v_t \sin(\gamma)\cos(\gamma) = l\dot{\theta}\cos(\gamma) + l_t\dot{\theta}_t\cos^2(\gamma)$

$(2.5)\cdot \sin(\gamma)$

$$v\sin(\gamma) = v_t \cos(\gamma)\sin(\gamma) + \dot{\theta}_t l_t \sin^2(\gamma)$$

$$\Rightarrow \quad l\dot{\theta}\cos(\gamma) + l_t\dot{\theta}_t\cos^2(\gamma) = v\sin(\gamma) - \dot{\theta}_t l_t \sin^2(\gamma)$$

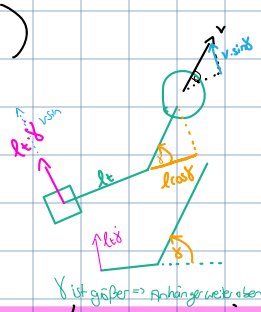
$$\Leftrightarrow \quad l\dot{\theta}\cos(\gamma) + l_t \underbrace{\dot{\theta}_t}_{=\dot{\theta}-\dot{\gamma}} = v\sin(\gamma) \qquad \gamma = \theta - \theta_t$$

$$\Leftrightarrow \quad \dot{\theta}\left(l\cos(\gamma) + l_t\right) = l_t\dot{\gamma} + v\sin(\gamma)$$

$$\Leftrightarrow \quad \dot{\theta} = \frac{l_t\dot{\gamma} + v\sin(\gamma)}{l\cos(\gamma) + l_t} \qquad (27)$$

Geschwindigkeitskomponente seitlich hinten

Hebelarm in Richtung Anhängerachse

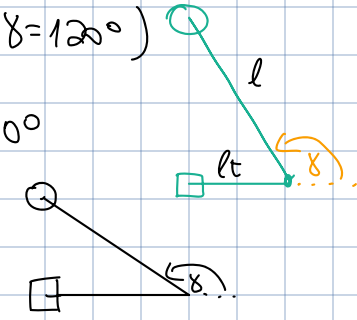


die seitliche Geschwindigkeit am Anhänger muss durch eine passende Drehung $\dot{\theta}$ des Roboters kompensiert werden (no lateral slip)

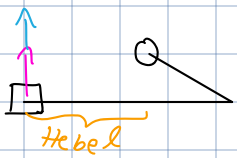
Singularität $l\cos\gamma + l_t = 0$ (tritt nicht auf bei input $\dot{\theta}$ statt γ s.u.)

(Fall $l_t = \frac{1}{2}l$ $\rightsquigarrow$ $l\cos\gamma + \frac{1}{2}l = 0 \Leftrightarrow \cos\gamma = -\frac{1}{2} \Rightarrow \gamma = 120°$)

(Fall $l_t = \underbrace{\frac{\sqrt{3}}{2}}_{\approx 0.86} l$ $\rightsquigarrow$ $l\cos\gamma + \frac{\sqrt{3}}{2}l = 0 \Leftrightarrow \cos\gamma = -\frac{\sqrt{3}}{2} \Rightarrow \gamma = 150°$



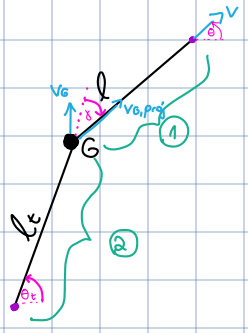
(Fall $l_t > l$ kein Problem



Singularität, wenn Roboter/Anhänger im Anhänger-KS die gleiche x-Koordinate haben

$$\text{Probot}_{(\text{trailer-KS})} = \begin{pmatrix} l_t \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} \cos\gamma\, l \\ \sin\gamma\, l \\ 0 \end{pmatrix} \rightsquigarrow 0 = l_t + \cos\gamma \cdot l$$

$\Rightarrow$ $\dot{\theta}$ kann nicht eindeutig bestimmt werden



$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} -l \\ 0 \end{pmatrix} + \begin{pmatrix} \cos\theta_t & -\sin\theta_t \\ \sin\theta_t & \cos\theta_t \end{pmatrix} \begin{pmatrix} -l_t \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} x - \cos(\theta) l - \cos(\theta_t) l \\ y - \sin(\theta) l - \sin(\theta_t) l \end{pmatrix}$$

Zustandsvektor $X = \begin{pmatrix} x \\ y \\ \theta \\ \gamma \end{pmatrix}$ $\quad U = \begin{pmatrix} v \\ \dot{\gamma} \end{pmatrix}$

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\gamma} \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ \frac{\sin\gamma}{l\cos\gamma + l_t} & \frac{l_t}{l\cos\gamma + l_t} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v \\ \dot{\gamma} \end{pmatrix}$$

$\gamma = \theta - \theta_t$

privilegierte Koordinaten

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\gamma} \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ \frac{\sin\gamma}{l\cos\gamma + l_t} & \frac{l_t}{l\cos\gamma + l_t} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v \\ \dot{\gamma} \end{pmatrix}$$

$l = l_t = 1$

$$X_1 = \begin{pmatrix} \cos x_3 \\ \sin x_3 \\ \frac{\sin x_4}{\cos x_4 + 1} \\ 0 \end{pmatrix} = \begin{pmatrix} \cos x_3 \\ \sin x_3 \\ \tan\left(\frac{x_4}{2}\right) \\ 0 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\cos x_4 + 1} \\ 1 \end{pmatrix}$$

$-\frac{\pi}{2} < \frac{x_4}{2} < \frac{\pi}{2}$

$-\pi < x_4 < \pi$

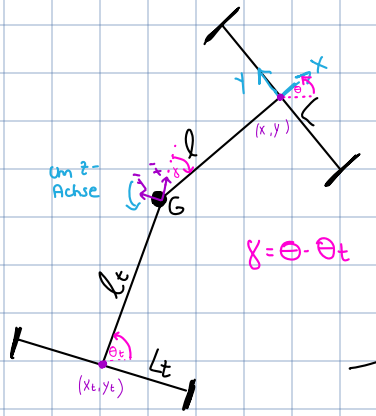
$\tan(\gamma) \overset{\text{abl.}}{\Rightarrow} \frac{1}{\cos^2\gamma}$

$\frac{1}{\cos^2\left(\frac{x_4}{2}\right)\frac{1}{2}} = \frac{2}{\cos^2\left(\frac{x_4}{2}\right)}$

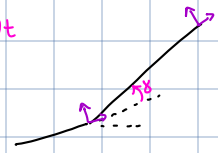
$$X_3 = [X_1, X_2] = \frac{\partial X_2}{\partial x} X_1 - \frac{\partial X_1}{\partial x} X_2$$

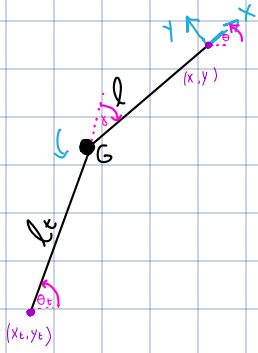
$$= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sin x_4}{(\cos x_4 + 1)^2} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \cos x_3 \\ \sin x_3 \\ \tan\left(\frac{x_4}{2}\right) \\ 0 \end{pmatrix} - \begin{pmatrix} * & * & -\sin x_3 & 0 \\ * & * & \cos x_3 & 0 \\ * & * & 0 & \\ * & * & 0 & \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\cos x_4 + 1} \\ 1 \end{pmatrix} \dots$$

eigene Modellierung Matlab



Wheels collapse in middle ⟹





$$x = \begin{pmatrix} x_c \\ y_c \\ \Theta \\ \gamma \\ \varphi \\ \varphi_t \end{pmatrix} \qquad v = \begin{pmatrix} \dot{\varphi} l \\ \dot{\gamma} \end{pmatrix} \text{ oder } \quad v = \begin{pmatrix} \dot{\varphi} l \\ \dot{\gamma} \end{pmatrix}$$

$\gamma = \Theta - \theta_t$

$\theta_t = \Theta - \gamma$

im Roboter-KS — im trailer-KS

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} \cos\Theta & -\sin\Theta \\ \sin\Theta & \cos\Theta \end{pmatrix} \begin{pmatrix} -l \\ 0 \end{pmatrix} + \begin{pmatrix} \cos\theta_t & -\sin\theta_t \\ \sin\theta_t & \cos\theta_t \end{pmatrix} \begin{pmatrix} -l_t \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} x - \cos(\Theta) l - \cos(\theta_t) l \\ y - \sin(\Theta) l - \sin(\theta_t) l \end{pmatrix}$$

im Chassis KS:

$$\begin{pmatrix} x_t \\ y_t \end{pmatrix} = \begin{pmatrix} -l \\ 0 \end{pmatrix} + \underbrace{\begin{pmatrix} \cos(-\gamma) & -\sin(-\gamma) \\ \sin(-\gamma) & \cos(-\gamma) \end{pmatrix}} \begin{pmatrix} -l_t \\ 0 \end{pmatrix} = \begin{pmatrix} -l - \cos(\gamma) l_t \\ \sin(\gamma) l_t \end{pmatrix}$$

$$= \begin{pmatrix} \cos(\gamma) & \sin(\gamma) \\ -\sin(\gamma) & \cos(\gamma) \end{pmatrix}$$

- no lateral slip (5.23 eines constraint)

$$v_{wn_NHC}(1) \sin\Theta - v_{wn_NHC}(2) \cos\Theta = 0$$

v_{wn} Θ_1 v_{wn} Θ_1 ~ NHC_wn v_{wn}

aus $\left\langle \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix}, \underbrace{\begin{pmatrix} \sin\Theta \\ -\cos\Theta \end{pmatrix}} \right\rangle = 0$

Richtung 90° zur Rollrichtung $\begin{pmatrix} \cos\Theta \\ \sin\Theta \end{pmatrix}$

- rolling without slipping (S. 23 zweites constraint)

$$\underbrace{v_{w1\text{-}NHC}(1)\cos\theta + v_{w1\text{-}NHC}(2)\sin\theta}_{t_1 \quad \theta_1 \quad t_1 \quad \theta_1} = \underbrace{\dot{\varphi}_1 R}_{t_1}$$

zurückgelegte Strecke mit Rad Bodenkontakt

$$\angle\, v_{w1\text{-}NHC}, \underbrace{\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}}_{\text{Radrichtung}} >$$

$\rightsquigarrow$ zu $=0$ umformen und benennen als NHC_w1_b

t_{w1}

- Write scleronomic implicit constraints as Pfaffian constraints

$$\phi_v(x,\dot{x},\cancel{t}) = 0 \quad \rightarrow \quad \phi_v(x,\dot{x}) = A(y)\,\dot{y}$$

$$\Rightarrow a_{1_w1} = \text{jacobian}(NHC_w1,\ \dot{y})$$

$$\vdots$$

$\Rightarrow$ Pfaffian constraint matrix

$$\underbrace{\begin{pmatrix} a_{1_w1} \\ \vdots \\ a_{2_tw1} \end{pmatrix}}_{=A_y} \dot{y} = 0$$

6×1

4×6

$$\text{rank}(A_y) = 4$$

$$A_y \cdot \underbrace{\dot{y}}_{=G(y)s} = 0$$

$\Rightarrow G(y)$ erhält man über den Nullraum von A

$\Rightarrow G(y)$ ist $6\times(6-4) = 6\times2$

$\Rightarrow$ nur 2 unabhängige inputs

andere Paper nehmen $v=\text{const.}$, oder (v,ω) von driver vorgegeben, ...

$$A_y = \begin{pmatrix} \sin\theta & -\cos\theta & 0 & 0 & 0 & 0 \\ \cos\theta & \sin\theta & 0 & 0 & -R & 0 \\ -\sin(\gamma-\theta) & -\cos(\gamma-\theta) & l_t + l\cos\gamma & -l_t & 0 & 0 \\ \cos(\gamma-\theta) & -\sin(\gamma-\theta) & l\sin\gamma & 0 & 0 & -R \end{pmatrix}$$

$A_y \cdot G = 0$

$$x = \begin{pmatrix} x_c \\ y_c \\ \theta \\ \gamma \\ \varphi \\ \varphi_t \end{pmatrix}$$

explicit Kinematics inputs: $\begin{pmatrix} \dot\varphi \\ \omega \end{pmatrix}$ oder $\begin{pmatrix} v \\ \omega \end{pmatrix}$

- Mittelpunkt der Achse $r_{m,p} = \begin{pmatrix} x_c \\ y_c \\ h \end{pmatrix} \xrightarrow[t]{\text{diff nach}} v_{m,p}$

- einsetzen in $v_{m,p}$ ergibt $v_{m,p,phi}$ tatsächliche Geschwindigkeit — hier taucht aber sowieso kein $\dot\theta$ auf

  aus Anschauung und mit $\dot\theta = \omega$ (da nur 1 Rad) → subs($v_{m,p}$, $\dot\theta$, $\underbrace{s_2}_{=\omega}$)

$$v_{m,p,NH} = \begin{pmatrix} \cos\theta\, v_p \\ \sin\theta\, v_p \\ 0 \end{pmatrix} = \begin{pmatrix} R\cos\theta\, \dot\varphi_e \\ R\sin\theta\, \dot\varphi_e \\ 0 \end{pmatrix}$$

es laubte nichtholonome Geschwindigkeit

$\Rightarrow v_{m,p,phi} = v_{m,p,NH}$

gewünschte Form: $\begin{pmatrix} \dot x_c \\ \dot y_c \\ \dot h \end{pmatrix} = G(y) \underbrace{\begin{pmatrix} \dot\varphi_1 \\ \dot\varphi_2 \end{pmatrix}}_{\text{generalized velocities } s}$

(sieht komplizierter aus als es ist :)

$$G = -\text{jac}\left(v_{m,p,phi} - v_{mp,NH}, \begin{pmatrix} \dot\varphi \\ \omega \end{pmatrix}\right) = \begin{pmatrix} R\cos\theta \\ R\sin\theta \\ 0 \end{pmatrix}$$

- $\dot\theta = \omega$

- S.O. :

$$\dot{\theta} = \frac{l_t \dot{\gamma} + v\sin(\gamma)}{l\cos(\gamma) + l_t}$$

$$\Leftrightarrow \quad \dot{\theta}\,(l\cos\gamma + l_t) = l_t\dot{\gamma} + v\sin(\gamma)$$

$$\Leftrightarrow \quad \frac{\dot{\theta}\,(l\cos\gamma + l_t) - v\sin(\gamma)}{l_t} = \dot{\gamma} = \begin{pmatrix} -\frac{\sin\gamma}{l_t} & \frac{l\cos\gamma + l_t}{l_t} \end{pmatrix}\begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$\downarrow \; v = \dot{\varphi}_l \cdot R \qquad \dot{\varphi}_l = \underbrace{\frac{1}{R}\, v}_{\text{Aw} \to \varphi}$$

$$\dot{\gamma} = \begin{pmatrix} -\frac{\sin\gamma}{l_t} R & \frac{l\cos\gamma + l_t}{l_t} \end{pmatrix}\begin{pmatrix} \dot{\varphi} \\ \omega \end{pmatrix}$$

- $\dot{\varphi}_{t1}$

$$v_t\sin(\gamma) = l\dot{\theta} + l_t\dot{\theta}_t\cos(\gamma) \quad (2.6)$$

$$v = v_t\cos\gamma + \dot{\theta}_t l_t \sin\gamma \quad (2.5)$$

$\gamma = \theta - \theta_t$

$\theta_t = \theta - \gamma$

$$\hookrightarrow \quad v_t = \frac{v - \dot{\theta}_t l_t \sin\gamma}{\cos\gamma} = \frac{v - (\dot{\theta} - \dot{\gamma}) l_t \sin\gamma}{\cos\gamma}$$

$$= \begin{pmatrix} \frac{1}{\cos\gamma} & -l_t\tan(\gamma) \end{pmatrix}\begin{pmatrix} v \\ \omega \end{pmatrix} + \underbrace{\dot{\gamma}}\, l_t\tan(\gamma)$$

$$\begin{pmatrix} -\frac{\sin\gamma}{l_t} & \frac{l\cos\gamma + l_t}{l_t} \end{pmatrix}\begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\cos\gamma} & -l_t\tan(\gamma) \end{pmatrix}\begin{pmatrix} v \\ \omega \end{pmatrix} + \begin{pmatrix} -\sin\gamma\tan\gamma & (l\cos\gamma + l_t)\tan\gamma \end{pmatrix}\begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\cos\gamma} - \sin\gamma\tan\gamma & l\cos\gamma\tan\gamma \end{pmatrix}\begin{pmatrix} v \\ \omega \end{pmatrix} = \dot{\varphi}_t \cdot R$$

$$\Leftrightarrow \quad \dot{\varphi}_t = \begin{pmatrix} \frac{1}{R}\left(\frac{1}{\cos\gamma} - \sin\gamma\tan\gamma\right) & \frac{l\cos\gamma\tan\gamma}{R} \end{pmatrix}\begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$\Rightarrow \dot{\varphi}_t = \begin{pmatrix} \frac{1}{\cos\gamma} & -\sin\gamma\tan\gamma & \frac{\ell\cos\gamma\tan\gamma}{R} \end{pmatrix} \begin{pmatrix} \dot{\varphi} \\ \omega \end{pmatrix}$$

Simplify $\Rightarrow$

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\gamma} \\ \dot{\varphi} \\ \dot{\varphi}_t \end{pmatrix} = \begin{pmatrix} R\cos\theta & 0 \\ R\sin\theta & 0 \\ 0 & 1 \\ -\frac{R}{\ell_t}\sin\gamma & 1+\frac{\ell\cos\gamma}{\ell_t} \\ 1 & 0 \\ \cos\gamma & \frac{\ell}{R}\sin\gamma \end{pmatrix} \begin{pmatrix} \dot{\varphi} \\ \omega \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \\ -\frac{1}{\ell_t}\sin\gamma & 1+\frac{\ell\cos\gamma}{\ell_t} \\ \frac{1}{R} & 0 \\ \frac{1}{R}\cos\gamma & \frac{\ell}{R}\sin\gamma \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix}$$

mit 2 Rädern

ähnlich aber für $\dot{\varphi}_t$:

$$v_t = \begin{pmatrix} \frac{1}{\cos\gamma} & -\sin\gamma\tan\gamma & l\cos\gamma\tan\gamma \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$v_{tw1} = v_t + \dot{\theta}_t (r_{tw} - r_T)^\perp \quad \text{(dabei in trailer-KS bleiben)}$$

$$= \left[\begin{pmatrix} v_t \\ 0 \end{pmatrix} + \dot{\theta}_1 \begin{pmatrix} 0 \\ \frac{L_t}{2} \end{pmatrix}^\perp \right] = \left[\begin{pmatrix} v_t \\ 0 \end{pmatrix} + \dot{\theta}_1 \begin{pmatrix} -\frac{L_t}{2} \\ 0 \end{pmatrix} \right]$$

$$= \begin{pmatrix} v_t - \frac{L_t}{2}\dot{\theta}_1 \\ 0 \end{pmatrix} = \begin{pmatrix} v_t - \frac{L_t}{2}(\dot{\theta} - \dot{\gamma}) \\ 0 \end{pmatrix}$$

$$\Rightarrow v_{tw1} = v_t - \frac{L_t}{2}\omega + \frac{L_t}{2}\dot{\gamma}$$

$$= \begin{pmatrix} \frac{1}{\cos\gamma} & -\sin\gamma\tan\gamma & l\cos\gamma\tan\gamma \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix} - \frac{L_t}{2}\omega$$

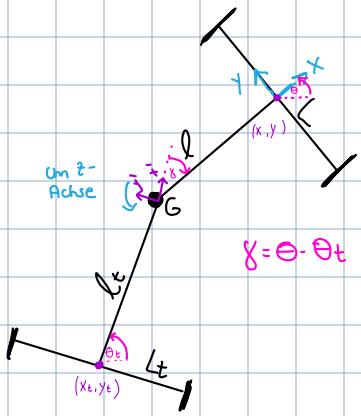
$$+ \frac{L_t}{2} \begin{pmatrix} -\frac{\sin\gamma}{l_t} & 1 + \frac{l\cos\gamma}{l_t} \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\cos\gamma} - \sin\gamma\tan\gamma - \frac{L_t}{2l_t}\sin\gamma & l\cos\gamma\tan\gamma \underbrace{- \frac{L_t}{2} + \frac{L_t}{2}}_{=0} + \frac{l\cos\gamma L_t}{l_t \cdot 2} \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$\Rightarrow \dot{\varphi}_{tw1} = \frac{v_{tw1}}{R} = (\dots \quad \dots) \frac{1}{R} \begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\gamma} \\ \dot{y}_t \end{pmatrix} = \begin{pmatrix} \frac{1}{2}R\cos\theta & \frac{1}{2}R\cos\theta \\ \frac{1}{2}R\sin\theta & \frac{1}{2}R\sin\theta \\ -\frac{R}{L} & \frac{R}{L} \\ -0.15\cos(\gamma - 0.17\sin\gamma) - 0.15) & 0.15\cos(\gamma - 0.17\sin\gamma) + 0.15) \\ 1 & 1 \\ -\sin\gamma & \cos\gamma - 0.18\sin\gamma \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \\ -10\sin\gamma & \cos\gamma + 1 \\ 2y & -3.4 \\ 2y\cos\gamma - 34\sin\gamma & 3.4\cos\gamma + 2y\sin\gamma \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix}$$

Dynamik



$\gamma = \theta - \theta_t$

$$x = \begin{pmatrix} x_c \\ y_c \\ \theta \\ \gamma \\ \varphi \\ \varphi_t \\ \dot{\varphi}_l \\ \dot{\varphi}_r \end{pmatrix} \quad \text{Rad-geschwindigkeiten} \qquad u = \begin{pmatrix} \tilde{\tau}_l \\ \tilde{\tau}_r \\ \tilde{\tau}_\gamma \end{pmatrix}$$

$\gamma = \theta - \theta_t$

$\theta_t = \theta - \gamma$

Momente:

$$l_{w1} = S_{pC}\, S_{C\tilde{w}_1} \begin{pmatrix} 0 \\ \tilde{\tau}_1 \\ 0 \end{pmatrix}$$ ← dreht um y-Achse Rad-KS

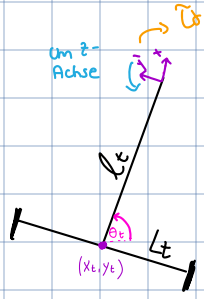
$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\phi_1 & 0 & \sin\phi_1 \\ 0 & 1 & 0 \\ -\sin\phi_1 & 0 & \cos\phi_1 \end{pmatrix}$$

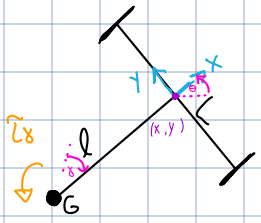
$$l_{w2} = S_{pC}\, S_{C\omega_2} \begin{pmatrix} 0 \\ \tilde{\tau}_2 \\ 0 \end{pmatrix}$$

$$l_C = S_{pC} \begin{pmatrix} 0 \\ 0 \\ \tilde{\tau}_3 \end{pmatrix}$$

$$l_t = S_{pC}\, S_{CT} \begin{pmatrix} 0 \\ 0 \\ -\tilde{\tau}_3 \end{pmatrix}$$

$$\begin{pmatrix} \cos\delta & \sin\delta & 0 \\ -\sin\delta & \cos\delta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$





$\gamma = \theta - \theta_t$

$\tilde{\tau}_\gamma$ wirkt auf

$\gamma = \theta - \theta_t$

Ergebnis: $M \begin{pmatrix} \ddot{\varphi}_e \\ \ddot{\varphi}_r \end{pmatrix} + k = q$

$$M = \begin{pmatrix} 2e\text{-}5 \cdot \cos(\gamma)^2 - 6e\text{-}5 \cdot \sin(2\gamma) + 1.2e\text{-}3 & 3e\text{-}4 - 1e\text{-}4 \sin(\gamma)^2 \\ * & 6e\text{-}5 \cdot \cos(\gamma)^2 + 1e\text{-}3 \end{pmatrix}$$ symm

$k = 0$

$$q = \begin{pmatrix} 1 & 0 & -0.15 - 0.73\,(0.2\cos\gamma + 0.24\sin\gamma) \\ 0 & 1 & 0.15 + 0.73\,(0.2\cos\gamma - 0.24\sin\gamma) \end{pmatrix} \begin{pmatrix} \tilde{\tau}_{w1} \\ \tilde{\tau}_{w2} \\ \tilde{\tau}_\gamma \end{pmatrix}$$

$= 0$ für $\gamma = 0$! kein Bu sondern $B(\gamma)u$

Überprüfung:

1) $\tau_\gamma = 0$, $l = 0$ $\Rightarrow$ dann wie normaler robot + trailer

$$\Rightarrow G_{y,rw} = \begin{pmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \\ -5.1\sin\gamma & 1 \\ 29 & -3.4 \\ 29\cos\gamma - 18\sin\gamma & 0 \end{pmatrix}, \quad G_y = \begin{pmatrix} 0.018\cos\theta & 0.018\cos\theta \\ 0.018\sin\theta & 0.018\sin\theta \\ -0.15 & 0.15 \\ -0.09\sin\gamma - 0.15 & 0.15 - 0.09\sin\gamma \\ 1 & 0 \\ 0.5\cos\gamma - 0.31\sin\gamma & 0.5\cos\gamma - 0.31\sin\gamma \end{pmatrix}$$

$G_{y,rw}$: identisch außer 4. Zeile

G_y: identisch zu trailer außer 4. Zeile

$$\dot{\gamma} = -5.1\sin\gamma\, v + \omega$$

$\dot{\gamma} = \dot{\theta} - \dot{\theta}_t$, $\omega = \dot{\theta}$

$$\dot{\gamma} = (-0.09\sin\gamma - 0.15)\dot{\varphi}_1 + (0.15 - 0.09\sin\gamma)\dot{\varphi}_2$$

$\dot{\gamma} = \dot{\theta} - \dot{\theta}_t$

identisch

$$\dot{\theta}_t = \dot{\theta} + (0.09\sin\gamma + 0.15)\dot{\varphi}_1 + (-0.15\cos\gamma + 0.09\sin\gamma)\dot{\varphi}_2$$

$$= 0.09\sin\gamma\,\dot{\varphi}_1 + 0.09\sin\gamma\,\dot{\varphi}_2$$

identisch

trailer: $\dot{\theta}_t = 5.1\sin(\theta - \theta_t)\, v$

trailer: $\dot{\theta}_t = 0.09\sin(\underbrace{\theta - \theta_t}_{=\gamma})\,\dot{\varphi}_1 + 0.09\sin(\underbrace{\theta - \theta_t}_{=\gamma})\,\dot{\varphi}_2$

$$\Rightarrow M = \begin{pmatrix} 1e\text{-}4\cos(\gamma)^2 + 1e\text{-}3 & 3.1e\text{-}4 - 1.4e\text{-}4\sin(\gamma)^2 \\ * & 1e\text{-}4\cos(\gamma)^2 + 1e\text{-}3 \end{pmatrix}$$

$l = 0$

$$q = \begin{pmatrix} 1 & 0 & -0.15 - 0.09\sin\gamma \\ 0 & 1 & 0.15 - 0.09\sin\gamma \end{pmatrix} \begin{pmatrix} \tau_{w_1} \\ \tau_{w_2} \\ \tau_\gamma \end{pmatrix}$$

M von robot + trailer sieht anders aus, aber $\cos^2(\gamma) = \frac{1+\cos(2\gamma)}{2}$

$$M = \begin{pmatrix} 7e\text{-}5\cos(2\gamma) + 1e\text{-}3 & * \\ 7e\text{-}5\cos(2\gamma) + 2e\text{-}4 & 7e\text{-}5\cos(2\gamma) + 1e\text{-}3 \end{pmatrix}$$

↗ robot + trailer

2) $\gamma = \theta - \theta_t = 0$, $\tau_{w1} = \tau_{w2} = 0$

$\Rightarrow M \begin{pmatrix} \ddot{\varphi}_e \\ \ddot{\varphi}_r \end{pmatrix}_{\underbrace{\dot{\gamma} = 0}} = q$ mit $q = \begin{pmatrix} -0.15\,\tau_\gamma \\ 0.15\,\tau_\gamma \end{pmatrix}$, $M = \begin{pmatrix} 2e\text{-}5 + 1.2e\text{-}3 & 3e\text{-}4 \\ * & 6e\text{-}5 \end{pmatrix}$

Simulation open-loop

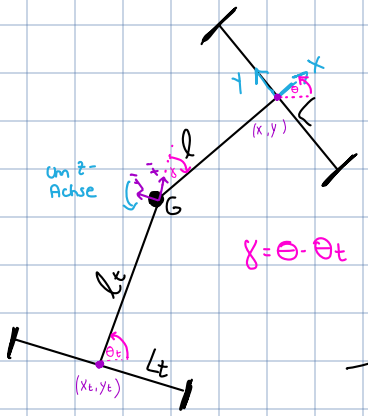
		gerade ausgeführt + Gelenkkalibrierung
$x =$	x_c	0
	y_c	0
	θ	0
	γ	0
	φ	0
	φ_t	0
	$\dot{\varphi}_e$	$\dot{\varphi}_{e,0}$
	$\dot{\varphi}_r$	$\dot{\varphi}_{e,0}$
$u =$	τ_e	0
	τ_r	0
	τ_γ	$\neq 0$

$M \begin{pmatrix} \ddot{\varphi}_e \\ \ddot{\varphi}_r \end{pmatrix}_{\underbrace{\dot{\gamma} = 0}} = q$

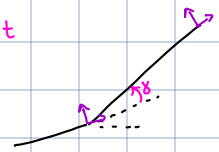
$\begin{pmatrix} \ddot{\varphi}_e \\ \ddot{\varphi}_r \end{pmatrix} = M^{-1} \underline{\underline{q(\gamma, \tau)}} = \underset{2\times2}{M^{-1}} \underset{2\times3}{B(\gamma)} \begin{pmatrix} \tau_{\varphi_1} \\ \tau_{\varphi_2} \\ \tau_\gamma \end{pmatrix}$

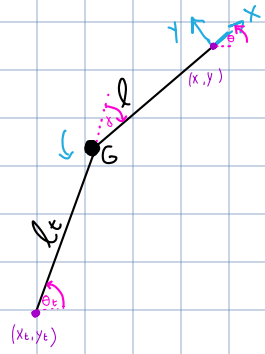
Dynamik mit collapsed wheels

$$X = \begin{pmatrix} x_c \\ y_c \\ \theta \\ \gamma \\ \varphi \\ \varphi_t \end{pmatrix} \qquad U = \begin{pmatrix} \dot{\varphi}_e \\ \dot{\gamma} \end{pmatrix} \text{ oder } \begin{pmatrix} v \\ \dot{\gamma} \end{pmatrix}$$



Wheels collapse in middle $\Rightarrow$





$\gamma = \theta - \theta_t$

$\theta_t = \theta - \gamma$

Kinematik

- $\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} v \qquad v = \dot{\varphi}_e R \qquad \begin{pmatrix} \dot{\varphi}_e \\ \dot{\gamma} \end{pmatrix} = \begin{pmatrix} \frac{1}{R} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v \\ \dot{\gamma} \end{pmatrix}$
- $\dot{\gamma} = \dot{\gamma}$
- $\dot{\varphi} = \dot{\varphi} = \frac{1}{R} v$

es fehlen: $\dot{\theta}, \dot{\varphi}_t$

1) $\dot{\theta} = \frac{\ell_t \dot{\gamma} + v\sin(\gamma)}{\ell\cos(\gamma) + \ell_t} = \begin{pmatrix} \frac{\sin\gamma}{\ell\cos\gamma + \ell_t} & \frac{\ell_t}{\ell\cos\gamma + \ell_t} \end{pmatrix} \begin{pmatrix} v \\ \dot{\gamma} \end{pmatrix}$

2) aus vorherigen Rechnungen:

$$\dot{\varphi}_t = \begin{pmatrix} \frac{1}{R}\left(\frac{1}{\cos\gamma} - \sin\gamma\tan\gamma\right) & \frac{\ell\cos\gamma\tan\gamma}{R} \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix}$$

$$\Rightarrow \dot{\varphi}_t = \begin{pmatrix} \frac{1}{R\cos\gamma} - \frac{\sin\gamma\tan\gamma}{R} + \frac{\ell\cos\gamma\tan\gamma}{R} \frac{\sin\gamma}{\ell\cos\gamma + \ell_t} & \frac{\ell\cos\gamma\tan\gamma\ \ell_t}{R\ (\ell\cos\gamma + \ell_t)} \end{pmatrix} \begin{pmatrix} v \\ \dot{\gamma} \end{pmatrix}$$

für Paper: $l_1 = l_2 = 1$

$$M(q) = \begin{pmatrix} \dfrac{\mu_{11,0} + \mu_{11,c}\cos(\gamma)}{(\cos(\gamma)+1)^2} & * \\ \dfrac{\sin\gamma\,(\mu_{12,s} + \mu_{12,sc}\cos\gamma)}{(\cos(\gamma)+1)^2} & \dfrac{\mu_{22,cc}\cos(\gamma)^2}{(\cos(\gamma)+1)^2} + \mu_{22,0} \end{pmatrix}$$

$\gamma = q_4$

negativ

$$a(q_4) = \frac{1}{\cos(q_4)+1}$$

Nilpotente Approximation

$$z_1 = \begin{pmatrix} 1 - \frac{1}{8} z_2^2 + \frac{1}{4} z_2 z_3 - \frac{1}{8} z_3^2 + \dots \\ 0 \\ -z_2 - \frac{1}{12} z_2^3 + \dots \\ -z_3 - \frac{1}{8} z_2^3 + \frac{1}{8} z_2^2 z_3 - \frac{1}{8} z_2 z_3^2 + \dots \end{pmatrix}$$

	$(0\ 0\ 0\ 0)$	$(0\ 2\ 0\ 0)$	$(0\ 1\ 1\ 0)$	$(0\ 0\ 2\ 0)$
$w(l) - w_1$	-1	$2-1=1$	$1+2-1=2$	$2\cdot 2-1=3$

$w(l) - w_3$	$(0\ 1\ 0\ 0)$	$(0\ 3\ 0\ 0)$
	$1-2=-1$	$3-2=1$

$w(l) - w_4$	$(0\ 0\ 1\ 0)$	$(0\ 3\ 0\ 0)$
	$2-3=-1$	$1\cdot 3-3=0$

$$z_2 = \begin{pmatrix} 0 \\ 1 \\ -\frac{1}{4} z_2^2 \\ -\frac{1}{4} z_2^2 \end{pmatrix}$$

$\to (0\ 0\ 0\ 0)$ $\quad w(2) - w_2 = -1$

$\to (0\ 2\ 0\ 0)$ $\quad w(2) - w_3 = 2-2 = 0$ $\quad \Rightarrow$ fällt weg!

$\to (0\ 2\ 0\ 0)$ $\quad w(2) - w_4 = 2-3 = -1$ $\quad \Rightarrow$ fällt nicht weg

Unterschied zur nilp. Approx. robot + trailer!

$\Rightarrow$ nilpotent approximation:

$$\dot{z} = \begin{pmatrix} 1 \\ 0 \\ -z_2 \\ -z_3 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 1 \\ 0 \\ -\frac{1}{4} z_2^2 \end{pmatrix}$$

Homogenitätsbedingung

$$f(\Lambda_2 x, \Delta_2 u) = \begin{pmatrix} \lambda^{s_1} u_1 \\ \lambda^{s_2} u_2 \\ -\lambda^{r_2} z_2 \lambda^{s_1} u_1 \\ -\lambda^{r_3} z_3 \lambda^{s_1} u_1 - \frac{1}{4}\lambda^{2r_2} z_2^2 \lambda^{s_2} u_2 \end{pmatrix} \overset{r_2=2,\ s_1=1,\ r_3=3,\ s_2=1}{=} \begin{pmatrix} \lambda^{s_1} u_1 \\ \lambda^{s_2} u_2 \\ -\lambda^{r_2} z_2 u_1 \\ -\lambda^{r_3} z_3 u_1 - \frac{1}{4}\lambda^{r_3} z_2^2 u_2 \end{pmatrix}$$

$= \mathrm{diag}(\lambda^{r_1}, \dots, \lambda^{r_{nx}})$ $\mathrm{diag}(\lambda^{s_1} \dots \lambda^{s_{nu}})$

$$\lambda^{\tilde{\tau}} \Lambda_\lambda f(x,u) = \lambda^{\tilde{\tau}} \begin{pmatrix} \lambda^{r_1} u_1 \\ \lambda^{r_2} u_2 \\ -\lambda^{r_3} z_2 u_1 \\ -\lambda^{r_4=3} (z_3 u_1 + \frac{1}{4} z_2^2 u_2) \end{pmatrix}$$

Lie Algebra Rank Condition (LARC)

$$\Delta = \{g_1, g_2, g_3, g_4\}$$

- $x_4 \neq a\cos\left(-\frac{l_2}{l_1}\right)$ (sonst durch Null teilen, Singularität)

$$A = \begin{pmatrix} \cos x_3 & 0 & \sin x_3\, a(x_4) & \left[-\sin(x_3 - x_4) - \sin(x_3)\right] a(x_4)^2 \\ \sin x_3 & 0 & -\cos x_3\, a(x_4) & \left[\cos(x_3 - x_4) + \cos(x_3)\right] a(x_4)^2 \\ \sin x_4\, a(x_4) & a(x_4) & -a(x_4) & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\det(A) = 1 \cdot \det \begin{pmatrix} \cos x_3 & \sin x_3\, a & [\dots] a^2 \\ \sin x_3 & -\cos x_3\, a & [\dots] a^2 \\ \sin x_4\, a & -a & 0 \end{pmatrix} = \dots$$

Matlab: $\det(A) = \frac{2}{(\cos(x_4)+1)^3} = \frac{2}{a(x_4)^3}$

Residuum

$$\underbrace{\bar{f}(z,u)}_{\text{original}} = \underbrace{h(z,u)}_{\text{approx}} + \underbrace{R(z,u)}_{\text{residuum}}$$

to show: $|R_i(\lambda\Delta z, \Delta u)| \leq M\lambda^{r_i+\tilde{\tau}+\eta} \quad \forall i \in \mathcal{L}_{1:n} \quad \forall z,u \text{ with } \|z\| \leq \delta, \|u\| \leq \delta$

$\forall \lambda \in (0,1], \eta \in \mathbb{R}^+$

$$R(z,u) = \bar{f}(z,u) - h(z,u) = \begin{pmatrix} \cos\left(\frac{1}{2}(z_2 - z_3)\right) \\ 0 \\ -2\tilde{a}\sin(z_2) \\ 2\sin\left(\frac{1}{2}(z_2 - z_3)\right) - 2\tilde{a}\sin(z_2) \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 1 \\ 1-2\tilde{a} \\ 1-2\tilde{a} \end{pmatrix} u_2$$

$$\tilde{a} = \frac{1}{\cos(z_2)+1}$$

$$- \left[\begin{pmatrix} 1 \\ 0 \\ -z_2 \\ -z_3 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 1 \\ 0 \\ -\frac{1}{4}z_2^2 \end{pmatrix} u_2 \right]$$

$$= \begin{pmatrix} \cos\left(\frac{1}{2}(z_2 - z_3)\right) - 1 \\ 0 \\ -2\tilde{a}\sin(z_2) + z_2 \\ 2\sin\left(\frac{1}{2}(z_2 - z_3)\right) - 2\tilde{a}\sin(z_2) + z_3 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 0 \\ 1-2\tilde{a} \\ 1-2\tilde{a} + \frac{1}{4}z_2^2 \end{pmatrix} u_2$$

$$|R_1(\lambda_2 z, \Delta_2 v)| = \left|\left(\cos\left(\tfrac{1}{2}\left(\lambda^{r_2} z_2 - \lambda^{r_3} z_3\right)\right) - 1\right) \lambda^{s_1} v_1\right|$$

$|\cos(x) - 1| = 2\left|\sin\left(\tfrac{x}{2}\right)\right|^2 \leq \frac{x^2}{2}$

$$\leq \frac{\left(\frac{1}{2}\left(\lambda z_2 - \lambda^2 z_3\right)\right)^2}{2} \lambda |v_1| = \frac{\frac{1}{4}\lambda^2 (z_2 - \lambda z_3)^2}{2} \lambda |v_1| \leq \frac{1}{8}\lambda^3 (z_2 + \lambda z_3)^2 |v_1|$$

$$\leq \frac{1}{8}\lambda^3 (z_2 + z_3)^2 |v_1| \leq \frac{1}{8}\lambda^3 (2\delta)^2 \delta = \frac{1}{4}\delta^3 \lambda^3 \overset{!}{\leq} M_1 \lambda^{\underbrace{r_1}_{=1} + \underbrace{\tau}_{=0} + \eta} \Rightarrow \eta \in (0, 2]$$

$|R_2(\lambda_2 z, \Delta_2 v)| = 0 \checkmark$

$$|R_3(\lambda_2 z, \Delta_2 v)| = \Big| \underbrace{\left(-\frac{2\sin(\lambda^{r_2} z_2)}{\cos(\lambda^{r_2} z_2) + 1} + \lambda^{r_2} z_2\right) \lambda^{s_1} v_1}_{①} + \underbrace{\left(1 - \frac{2}{\cos(\lambda^{r_2} z_2) + 1}\right) \lambda^{s_2} v_2}_{②} \Big|$$

① $\frac{\sin(x)}{\cos(x) + 1} = \tan\left(\frac{x}{2}\right)$

$$\left(-2\underbrace{\tan\left(\tfrac{1}{2}\lambda z_2\right)}_{\tan(x) = \frac{x}{2} + \frac{x^3}{24} + \dots} + \lambda z_2\right) \lambda v_1 \leq -\underbrace{C_1}_{\in \mathbb{R}} \lambda^4 z_2^3 v_1$$

aber auf $|z_2| \leq \delta$ kann es nicht unendlich anwachsen & Schranke kann gefunden werden

② $\left(1 - \frac{2}{\cos(\lambda z_2) + 1}\right) \lambda v_2 \leq -\underbrace{C_2}_{\in \mathbb{R}} \lambda^2 z_2^2 \lambda v_2 = -C_2 \lambda^3 z_2^2 v_2$

$\frac{2}{\cos(\lambda z_2) + 1} = \frac{1}{2} + \frac{x^2}{8} + \dots$

gleiches Argument wie oben: kann nicht ∞ anwachsen $\forall |z_2| \leq \delta, \lambda \in (0, 1]$

gesamt: $|R_3(\dots, \dots)| \leq |-C_1 \lambda^4 z_2^3 v_1 - C_2 \lambda^3 z_2^2 v_2|$

$$\leq C_1 \lambda^4 \delta^3 + C_2 \lambda^3 \delta^3 \overset{!}{\leq} M_3 \lambda^{\underbrace{r_3}_{=2} + \tau + \eta} \Rightarrow \eta \in (0, 1]$$

$$|R_4(\lambda_4 z, \lambda_2 v)|$$

$$= \Big| \underbrace{\Big(2\sin\Big(\tfrac{1}{2}\big(\lambda^{r_2} z_2 - \lambda^{r_3} z_3\big)\Big) - \frac{2\sin(z_2)}{\cos(2z_2)+1} + \lambda^{r_3} z_3\Big)\lambda^{s_1} u_1}_{①} + \underbrace{\Big(1 - \frac{2}{\cos(\lambda^{r_2} z_2)+1} + \frac{1}{4} z_2^2\Big)\lambda^{s_2} u_2}_{②} \Big|$$

① $\dfrac{\sin(x)}{\cos(x)+1} = \tan\left(\dfrac{x}{2}\right)$

$$2\sin\left(\tfrac{1}{2}x\right) - x = -\frac{x^3}{24} + \frac{z^5}{1920} - \frac{z^7}{322\,560}$$

$$\Big|\Big(2\sin\Big(\tfrac{1}{2}\big(\lambda z_2 - \lambda^2 z_3\big)\Big) - \frac{2\sin(\lambda z_2)}{\cos(\lambda z_2)+1} + \lambda^2 z_3\Big)\lambda u_1\Big|$$

$$= \Big|\Big(2\sin\Big(\tfrac{1}{2}\big(\lambda z_2 - \lambda^2 z_3\big)\Big) - (\lambda z_2 - \lambda^2 z_3) + (\lambda z_2 - \lambda^2 z_3) - 2\tan\Big(\frac{\lambda z_2}{2}\Big) + \lambda^2 z_3\Big)\lambda u_1\Big|$$

$$\le \Big|\Big(\frac{\left|\frac{1}{2}(\lambda z_2 - \lambda^2 z_3)\right|^3}{24} + \lambda z_2 - 2\tan\Big(\frac{\lambda z_2}{2}\Big)\Big)\lambda u_1\Big|$$

$\tan(x) = \frac{x}{2} + \frac{x^3}{24} + \dots$

$$\le \Big|\frac{\left|\frac{1}{2}(\lambda z_2 - \lambda^2 z_3)\right|^3}{24}\lambda u_1\Big| + \Big|C_3 \lambda^3 z_2^3 \lambda u_1\Big|$$

$$\le \dots \lambda^4 \dots \quad \lambda^5 \dots \le M_4 \lambda^{\underbrace{r_4}_{=3} + \underbrace{\tau}_{=1} + \underbrace{n}_{=1}}$$

② $\Big|\Big(1 - \frac{2}{\cos(\lambda z_2)+1} + \frac{1}{4}\lambda^2 z_2^2\Big)\lambda^{s_2} u_2\Big|$

ohne 3. Term wie bei R_3:

$$\Big|\Big(1 - \frac{2}{\cos(\lambda z_2)+1}\Big)\lambda u_2\Big| \le \Big|\underbrace{-C_2}_{\in \mathbb{R}} \lambda^2 z_2^2 \lambda u_2\Big| = \Big|-C_2 \lambda^3 z_2^2 u_2\Big| \overset{!}{\le} M_4 \lambda^{\underbrace{r_4}_{=3} + \tau + n}$$

$\Rightarrow n = 0$

muss aber positiv sein!

$\frac{2}{\cos(\lambda z_2)+1} = \frac{1}{2} + \frac{x^2}{8} + \dots$

gleiches Argument wie oben: kann nicht ∞ anwachsen da $|z_2| \le 5$, $\lambda \in (0,1]$

mit 3. Term:

$$\left|\left(1 - \frac{2}{\cos(\alpha z_2)+1} + \frac{1}{4}\alpha^2 z_2^2\right)\alpha v_2\right| \leq \left|\frac{\alpha^4 z_2^4}{24}\alpha v_2\right| \leq \left|\frac{1}{24}\delta^5 \alpha^4\right|$$

$$\dot{\leq} M_4 \alpha^{\underbrace{r_4}_{=3} + \underbrace{\tau}_{=0} + n}$$

$$\Rightarrow n \in (0, 1]$$

$= \frac{1}{2} + \frac{x^2}{8} + \frac{x^4}{48} + \cdots$

gleiches Argument wie oben: kann nicht 0 annehmen da $|\alpha z_2| \leq \delta, \alpha \in (0, 1]$